



Zombie fires in the Arctic

Fire-resistant vegetation is burning in the Arctic, which doesn't bode well for global warming. <http://digit.in/oct20-31>



Nutrient-rich Plant droplets

Small droplets on the edges of blueberry bush leaves are loaded with nutrients for many insects. <http://digit.in/oct20-32>

Climate History

Sticking a thermometer into the past

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ssential to our understanding of how human activities are affecting the climate, is to look

back and see the history of climate to identify any long term trends. Geologists have found rocks associated with glaciers in the sands of Namibia, indicating ice sheets once covered the arid deserts there. Fossilised crocodiles and palms have been found in the arctic regions, showing that in the past, the polar regions once had tropical temperatures. Drilling samples of ice cores allows researchers to directly analyse small bubbles of air trapped within the ice, which contain clues about the ambient temperatures at that time. Temperature-related information can also be gleaned from the careful study of fossils, which have trapped within them isotopes that degrade at different rates depending on the temperatures at the time of fossilisation. Another approach is to drill deep into the crust and just stick thermometers in rocks. Rocks closer to the surface change

their temperatures much faster than rocks deep within the crust. By precisely and carefully measuring the change in temperature, it is possible to figure out how warm the planet was when the rocks were formed. These approaches all vary in granularity and period. The fossil crocodiles in the arctic give only a vague idea of the past temperatures. The ice cores give precise information but are only valuable for 1 million years. The data from all these various sources has to be churned together to produce a temperature map over time. This is further helped by astronomers working the orbital characteristics of the planet, and how that affects the climate. These variations are called Milankovitch cycles and the reconciliation of geological data with orbital characteristics is called astrochronology.

One of the biggest clues comes from phytane, a degraded form of chlorophyll, which is used by photosynthesizing plants for hundreds of millions of years. It is stable over incredibly long periods, and has been found in

marine fossils around the world. By correlating the concentration of phytane with carbon dioxide levels in the atmosphere, researchers have managed to piece together a 500 million year climate history of the planet.

Last month, researchers from UC Santa Cruz used the evidence from the shells of marine fossils to piece together a high fidelity record of the past 66 million years. The study was conducted on core samples extracted from seafloors. The evidence shows that the Earth moves through four phases, hothouse, warmhouse, coolhouse and icehouse. This is the highest resolution record of climate variability, and it shows that in a business as usual scenario, the temperatures of the planet can rise to the extremes seen during the hothouse phase. In other words, the human impact on climate change is projected to exceed that of orbital variations. We humans are pumping carbon dioxide into the atmosphere far faster than the natural processes that caused the most extreme periods of global warming the planet has seen so far.

HOTHOUSE

There have been two major spikes in global temperatures over the past hundred years, one about 94 million years ago, and another about 50 million years ago. These were the periods when the Earth was going through the hothouse phase. These two are called the Cretaceous Hot Greenhouse or the Cretaceous Thermal Maximum (CTM) and the Paleocene-Eocene Thermal Maximum (PETM) or the Late Paleocene Thermal Maximum.

There were a number of reasons that led to the CTM. The supercontinent of Pangea began breaking up between 150 to 130 million years ago. All the current continents were clubbed together in a single landmass around the south pole. Giant tears opened up along the ocean floors, heating the planet and giving rise to an increase in volcanic activities. The volcanos pumped carbon dioxide into the atmosphere, which trapped the